

RESEARCH ARTICLE

Drug and Dosage Recommendation Based on Explainable Generative AI Using Patient-Specific Modeling

EVREN DAGLARLI¹, (Senior Member, IEEE)

Faculty of Computer and Informatics Engineering, Istanbul Technical University, Maslak, 34469 Istanbul, Türkiye

e-mail: evren.daglarli@itu.edu.tr

This work was supported by Cognitive Systems Laboratory, Istanbul Technical University.

ABSTRACT Despite advances, personalized medicine involving patient-specific drug selection and dosing remains challenging. In this paper, we propose an interpretable, generative AI framework for drug recommendation and dosing. While clinical decisions often follow general guidelines, our approach uses patient-specific data to generate evidence-supported candidate drug-and-dose recommendations informed by up-to-date, provenance-linked sources. The Retrieval-Augmented Generation (RAG) framework combines data from electronic health records (EHRs) and employs machine learning and explainable AI methods. The digital twin approach supports this by simulating counterfactual treatment effects in a virtual patient model. Hence, the approach provides not only patient-tailored recommendations but also a clear, evidence-supported justification for the suggested drug and dose. We demonstrate feasibility via illustrations and simulations and outline record-based evaluations across multiple cohorts, reporting lower dose-prediction error, better calibration, and better discrimination of adverse drug reactions (AUC-ROC) and lower simulated ADR-risk indicators ($RRI = 0.78$; 95% CI 0.72–0.85) relative to guideline-based baselines, together with clinician-trust assessments. Retrospective results were consistent across heterogeneous cohorts, suggesting robustness to population and data shifts. The dataset comprises 75% real anonymized EHR records and 25% validated synthetic samples used only for training, with all test metrics computed on real-only data. We evaluate our approach on a de-identified, multi-cohort EHR-pharmacogenomics dataset of 20,030 patients spanning Hypertension, Diabetes Mellitus, Oncology, and Renal Impairment cohorts (hereafter, the ADR-20K multi-cohort dataset), constructed from anonymized EHRs, PharmGKB resources, and GAN-augmented synthetic samples with patient-level splits and temporal blocking to prevent leakage.

INDEX TERMS Drug recommendation, dosage control, explainable AI, generative AI, patient specific modeling.

I. INTRODUCTION

Personalized drug dosage optimization remains one of the most challenging tasks in clinical decision-making. Despite advances in computational medicine and machine learning, most existing systems either rely solely on structured patient data or lack transparency in their recommendations. This limitation can lead to suboptimal treatment outcomes, increased adverse drug reactions (ADRs), and reduced clinician trust in AI-assisted healthcare tools [1]. To address these

limitations, precision-medicine approaches leverage genomics, biomedical informatics, and AI to tailor therapy to each patient, replacing “one-size-fits-all” policies with individualized decisions [2]. By considering patient data, including genomic variants, biomarkers, and lifestyle factors, custom medicines and dosing can be tailored to maximize treatment effects, minimize side effects, and ultimately increase patient satisfaction and quality of life [3].

The basic feature of personalized medicine is its major impact on transforming healthcare provision, thereby changing the focus from empirical treatment guidelines to targeted therapies that fit patients’ unique biological and physiological

The associate editor coordinating the review of this manuscript and approving it for publication was Fabrizio Messina¹.

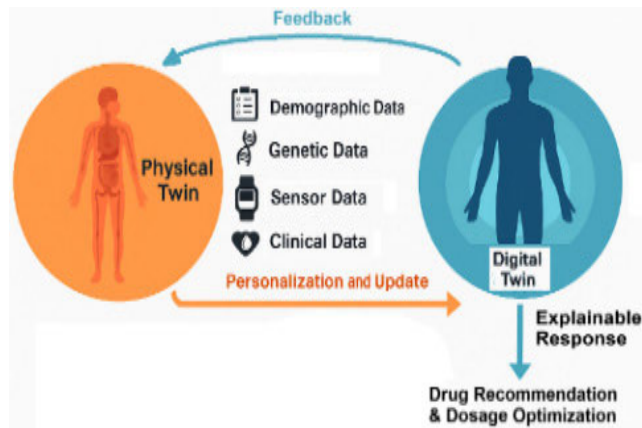


FIGURE 1. Digital-twin-based patient specific model and explainable AI interpretation.

profiles [4], [5]. This method optimizes therapeutic interventions and addresses the healthcare inefficiency problem caused by adverse drug reactions (ADRs), one of the primary issues in conventional medical practice. Despite substantial progress, patient-level drug selection and dosing remain under-specified due to heterogeneity and limited prospective interpretability; this gap calls for methods that couple evidence retrieval with calibrated, explainable dosing decisions [5]. The heterogeneity of patient populations, coupled with the complex interplay between their genes, environment, and diseases, makes it more challenging to gain a comprehensive understanding of the mode of action of drugs in individual biological systems. Primarily, trials must adhere to generalized principles and clinical testing that, unfortunately, are not always representative of the variability within the patient population. Along these lines, the demand for innovative computational methods that can leverage vast amounts of data and provide precise, patient-specific insights is increasing [6].

To address these gaps and challenges faced by the healthcare sector, we present a novel solution (Figure 1) involving an explainable generative-AI framework for patient-specific drug and dosage recommendation that integrates retrieval-augmented evidence access, a patient digital-twin simulator, and transformer-based learning within a single pipeline. The primary contribution of our work is the development of transparent and interpretable decision-making models that integrate deep learning with XAI, enabling physicians to understand the rationale behind recommended drug therapies and dosages and narrowing the gap between AI-enabled recommendations and clinical acceptance. Specifically, we unify a patient-specific model with retrieval-augmented conditioning, digital-twin simulation, and transformer-based learning to generate dosage recommendations together with calibrated ADR-risk estimates and case-level explanations. These models not only suggest candidate drug therapies and dosages but also provide evidence-supported explanations that clinicians can audit. By elevating transparency and understanding, our solution

aims to be the missing link between AI-enabled recommendations and clinical acceptance, thereby allowing clinicians to be more confident and achieve better patient care outcomes.

We next detail the system architecture and evaluation protocol, then present comparative results, limitations, and ethics/data-governance considerations for investigating the potential of explainable generative AI in personalized medicine in the context of drug and dosage recommendations. Through the exposition of a detailed methodology and the presentation of its operationalization in the form of case studies, we aim to demonstrate the practicality and advantages of our approach in enhancing clinical practice, thereby improving treatment accuracy and patient safety.

II. RELATED WORK

The healthcare industry has made huge strides in personalized medicine, which is mostly preferred due to the successful integration of informatics, AI, and computational biology. These have been the major driving forces for the health sector. This section of the paper explores the topic of drug recommender systems, accompanied by an explanation of the basic paradigm of generative AI and the role of explainable AI in the medical field.

A. DRUG RECOMMENDATION SYSTEMS

In the past, datasets collected through machine learning and statistical models have been utilized in drug recommendation systems, which serve as the foundation for these systems. The purpose of these systems is to make the doctor's task of finding the best treatment for the patient easier, and hence, recovery would be achieved. Collaborative filtering techniques utilize the concept of patient similarity, which involves establishing medical history and treatment response, to recommend therapies [7]. Additionally, knowledge-based systems utilize clinical practice guidelines and pharmacological databases as sources from which they derive patient-specific therapies tailored to local disease characteristics [8]. Traditional methods certainly have their weaknesses. They can, however, obtain an accurate human personal profile that covers genetic diversity, diseases, lifestyle, and other factors. In addition, when making decisions, depending only on suggestions from population-based studies restricts the adaptability of these methods if healthcare situations change unexpectedly.

B. GENERATIVE AND EXPLAINABLE AI TECHNIQUES IN HEALTHCARE

Variational Auto Encoders (VAEs) and Generative Adversarial Networks (GANs) are the primary tools of generative AI, a top technology for advancing personalized medicine through the generation of synthetic data and the discovery of deep patterns in patient-specific datasets [9], [10]. Such models enable the simulation of drug effects and the identification of the optimal dose by combining various patient cases that are not yet covered in clinical trials or medical databases. The implementation of explainable Artificial

Intelligence (XAI) frameworks is one way to address the interpretability and trust challenges related to AI-supported decisions in healthcare [11], [12]. XAI instruments, such as feature importance and rule extraction, enable physicians to understand the rationale behind AI predictions and verify whether the given solution is accurate [13]. This kind of transparency not only lays the foundation for better clinical decision-making but also changes the mindset of the medical community regarding the integration of AI into practice [14].

C. CRITIQUE OF CURRENT METHODOLOGIES AND GAPS IN THE LITERATURE

In some areas, development has been very efficient; however, the authors of the reviewed papers have also highlighted some limitations of existing works and omissions in the literature. Also, if we consider, for example, current medication suggestion systems, we will see that they are not sufficiently robust and may fail in the presence of sparse and noisy patient data which leads to the situation when, not only, treatment and other agents' use are suboptimized, but, also, there is the risk of adverse drug reactions (ADR) occurrence [15]. Moreover, the challenge of AI interpretability remains a stumbling block for the adoption of AI by clinicians who want the algorithms to be transparent in both clinical and regulatory contexts [16], [17].

Furthermore, it would be beneficial to emphasize that the available research studies often limit themselves to a single data source, such as genomics or clinical records [18], [19], [20]. These studies have not even remotely determined the degree of openness in multimodal data integration, which involves combining imaging data and patient-reported outcomes to achieve a deeper and more accurate understanding of disease mechanisms and treatment options [21].

So, in brief, on one hand, a lot of work has been done in the field of AI-powered personalized medicine, but on the other hand, there is still an enormous gap that calls for the development of advanced methods that are capable of interweaving different patient data in a seamless manner, making a data-driven analysis understandable and thereby giving clinicians the opportunity to select the best therapeutic option [22], [23], [24]. This publication addresses these problems by presenting its novel framework, an explanatory model of generative AI, designed to overcome the gaps in the field and enable the delivery of the most effective and personalized patient treatment.

Recent lines of research include (i) guideline calculators and rule-based systems, (ii) classical dosing regressors, (iii) RL-based treatment policies, (iv) LLM-assisted recommenders, (v) pharmacokinetic “digital-twin” simulators, and (vi) RAG-augmented clinical summarizers. Compared with these, our pipeline uniquely combines a digital-twin simulator with RAG-conditioned evidence ingestion and a transformer-based RL policy while providing auditable SHAP explanations and calibration safeguards (ECE). This combination yields patient-level personalization with

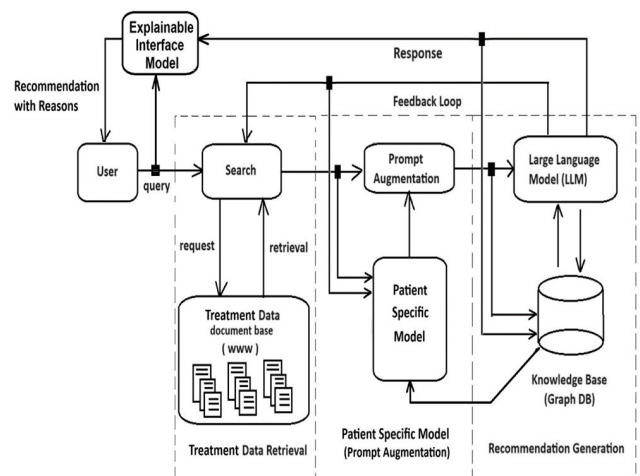


FIGURE 2. Explainable RAG-based drug and dosage recommendation system with patient-specific model.

quantified uncertainty and clinician-facing rationales rather than text-only advisories or non-RL recommenders.

III. METHODOLOGY

In this study, we develop a hybrid solution that incorporates several key elements. Such elements include a patient-specific treatment data retrieval mechanism and a generative AI framework for explaining the drug dosage recommendation application depicted in Figure 2, which is designed for a digital twin-based system. This line primarily describes a RAG framework with a deep learning-driven, explainable knowledge base, which is the principal structure of the system. According to the degree of similarity between the user prompt and the data from the explainable knowledge base, the retrieval mechanism in the RAG framework selects the relevant information concerning patient-specific treatment data.

A. TREATMENT DATA RETRIEVAL MECHANISM

This component, which is a crucial part of the RAG architecture, utilizes a dynamic database query system to extract relevant patient medical records, including historical prescriptions, treatment outcomes, and diagnostic data. Machine learning techniques, such as natural language processing (NLP), are integrated to interpret unstructured data from medical records and convert it into structured, actionable insights. By transforming narrative notes and diverse record formats into a uniform, query-ready information set, the system facilitates a comprehensive view of each patient's medical history. It ensures that even subtle clinical nuances are captured for subsequent analysis and review. When a user submits a query, the search component initially captures the raw data, which is then analyzed to identify and select the most pertinent data sources for processing. The processing function $r = \phi(qu, f)$ involves tokenization, normalization, named entity recognition (NER), and mapping to controlled

vocabularies or ontologies (e.g., UMLS or SNOMED-CT). Where $\phi: Q \rightarrow R$ is the transformation function that maps the user query to a formalized internal request $r \in R$, and $qu, f \in Q$ are a user-generated query in natural language and feedback response from the LLM, where Q denotes the space of all possible input queries.

The query engine constructs and sends custom queries that combine one set of patient-specific parameters, such as current symptoms, historical treatment responses, demographic details, and laboratory results, to collect the necessary information from the web. As soon as a query is launched in the form of a vectorized request, the system responds with a vast and well-updated repository that accesses an extensively indexed treatment data resource, which is continually expanded. The source of this resource is a high-quality dataset, which is a conglomerate of EHRs, electronic health record-based datasets of clinical trials, hospital information systems, and peer-reviewed literature. The repository that stores these data not only comprises a log of patient-reported symptoms and clinical observations but also contains comprehensive medication protocols, which provide target drugs, dosage information, and frequency guidelines. The system, thus, through the web crawler module, incorporates the treatment data document base, as well as sources of past treatment outcome records and diagnostic data, such as laboratory results and imaging studies, to provide a holistic view of patient care. The retrieved documents from the treatment data document base are thus sent back to the search component.

The retrieval function $D' = \psi(r, D)$ is a keyword-based method (BM25). At the same time, it also uses dense embedding similarity (cosine distance in a vector space) as a hybrid model that incorporates both lexical and semantic aspects. Where $\psi: R \times D \rightarrow D' \subseteq D$ is the document retrieval function that filters out relevant treatment data documents D' given the request r . Once the data is fetched, advanced machine learning techniques are applied to validate, normalize, and analyze the obtained information. Natural language processing methods identify key clinical entities and their relationships in the source texts, while the application of various classification and clustering techniques facilitates the division of treatment outcomes and correlation with patient profiles. This processing layer not only improves data accuracy but also identifies potential drug interactions or contraindications by comparing the current symptom picture with records. Rigorous filtering protocols ensure that only current and accurate pieces of evidence are included in the clinical decision-making toolbox, as outdated and irrelevant information is discarded.

The meticulously collected data is subsequently incorporated into a clinical decision support system of a broader scope, where it is matched with patient profiles and the latest clinical guidelines, enabling the system to provide physicians with actionable insights. Following personalized treatment recommendations may be formulated as such as altered medication doses, alternative drugs more consistent with the case, or innovative therapy proposals based on the latest research. To test this ongoing search method, a robust experimental

design was employed. Characteristics monitored very precisely in this experiment included data retrieval accuracy, processing time, and the clinical impact—patient satisfaction and treatment efficacy. The system's resiliency was additionally corroborated by detecting concordances with clinical datasets and along with quality requirements.

Utilizing a live query mechanism in conjunction with machine learning and an extensive, reliable data pool, the system for retrieving patient-specific treatment data significantly enhances the precision and customization of clinical recommendations. This is an achievement in the endeavor to obtain medical records of high complexity, which enables the generation of a more flexible and near-instantaneous treatment plan—the groundwork of the new method. The ongoing system's efficiency has been enhanced by the current improvements and the integration of new data sources and NLP, resulting in improved patient health outcomes and a more efficient healthcare system.

B. PATIENT-SPECIFIC MODEL AND DIGITAL TWIN-BASED EXPLAINABLE GENERATIVE AI FRAMEWORK

The conceptual AI framework based on digital twin technology represents a development in personalized medicine, enabling the creation of patient-specific simulations of drug interactions and treatment outcomes. The digital twin is the central component of the entire concept. It is a virtual model of the patient's physiological and clinical profile, created by combining data from various sources, including electronic health records (EHRs), genomic sequences, real-time biometric data from wearable devices, and patient-reported outcomes (PROs). Such a model functions like a patient simulator, facilitating experiments with various therapeutic interventions under both fixed and changing conditions. As a result, medical professionals can choose the optimal therapy and tailor the dosage with great accuracy.

1) ARCHITECTURAL COMPONENTS AND GENERATIVE AI INTEGRATION (DIGITAL-TWIN: PATIENT-SPECIFIC MODEL)

The framework utilizes AI generative models to create and extend specific scenarios for patients, thereby eliminating the need for static clinical guidelines. The suggested structure of the framework, illustrated in Figure 3, is that the patient-specific model is an advanced architecture that seamlessly integrates the Transformer-based reinforcement learning with generative augmentation strategies to enable personalized and context-aware decision-making. Initially, three crucial information streams—namely Search Output, Knowledge-Base Output, and Feedback Loop—are provided as parallel inputs into a Transformer Encoder module. The Transformer Encoder, owing to its multi-head self-attention mechanism, effectively captures complex interdependencies among these diverse information sources, yielding a comprehensive embedding representation denoted as E_{input} . This embedding consolidates real-time treatment data retrieval, historical domain knowledge, and experiential user feedback into a unified, dense latent space, ensuring that the model

possesses a nuanced understanding of the patient's dynamic context.

Algorithm 1 Patient-Specific Model

```

1: procedure sub_procedure_declaration (S_out, KB_out, F_loop)
2:    $E_{input} \leftarrow \text{TransformerEncoder}([S\_out, KB\_out, F\_loop])$ 
3:    $X_{synthetic} \leftarrow \text{GAN\_Generator}(E_{input}, \text{noise\_vector})$ 
4:    $E_{enhanced} \leftarrow \text{Concat}(E_{input}, \text{TransformerEncoder}(X_{synthetic}))$  #1
5:   while not done
6:      $\pi(at | E_{enhanced}) \leftarrow \text{TransformerPolicy}(E_{enhanced})$  #2
7:      $a^* = \text{argmax } \pi(at | E_{enhanced})$  #3
8:      $V(E_{enhanced}) = \text{TransformerValueEstimator}(E_{enhanced})$ 
9:      $\text{new\_state, reward, done} \leftarrow \text{EnvironmentSimulation}(\text{state}, \text{action})$ 
10:     $\text{Transformer\_RL.UpdatePolicy}(\text{state}, \text{action}, \text{reward}, \text{new\_state})$ 
11:     $\text{state} \leftarrow \text{new\_state}$ 
12:  end while
13: end procedure

14: #1: patient_embeddings
15: #2: action_probabilities
16: #3: optimal action

```

Following the initial embedding generation, E_{input} is fed into a GAN (Generative Adversarial Network) Generator alongside a stochastic noise vector to generate synthetic patient data that mirrors real-world variability, capturing latent factors such as genetic polymorphisms, metabolic rates, and environmental influences. The GAN Generator synthesizes realistic yet novel patient-like data samples, encapsulated as $X_{synthetic}$. These synthetic profiles are further passed through a second Transformer Encoder to project them into the same latent feature space, thereby ensuring consistency in representation. The results from the original embedding and the synthetically enhanced embeddings are concatenated, creating a more comprehensive and diversified feature set, which is referred to as $X_{enhanced}$. Such synthetic data is crucial for training models that operate correctly, particularly in scenarios where data sparsity or privacy constraints are present. This enrichment step is essential for addressing the sparsity or incompleteness issues that frequently occur in real-world clinical data, particularly when dealing with rare diseases or individual treatment histories.

Then, $X_{enhanced}$ plays the role of a new input to the Transformer Policy network, which in turn implements a policy for deciding on the extended latent space. The Transformer Policy leverages the temporal and relational structure inherent in the sequence of patient states and clinical actions, resulting in a probability distribution over the set of possible actions, namely action_probs . This distribution reflects the model's decisions about the most suitable interventions, given the patient's current state and the model's assumptions about the patient's future evolution. The action with the most significant probability is selected here using the argmax operation; thus, the most optimal action, which represents the system's last decision on the recommended treatment or clinical pathway, is chosen. By working with sequential data,

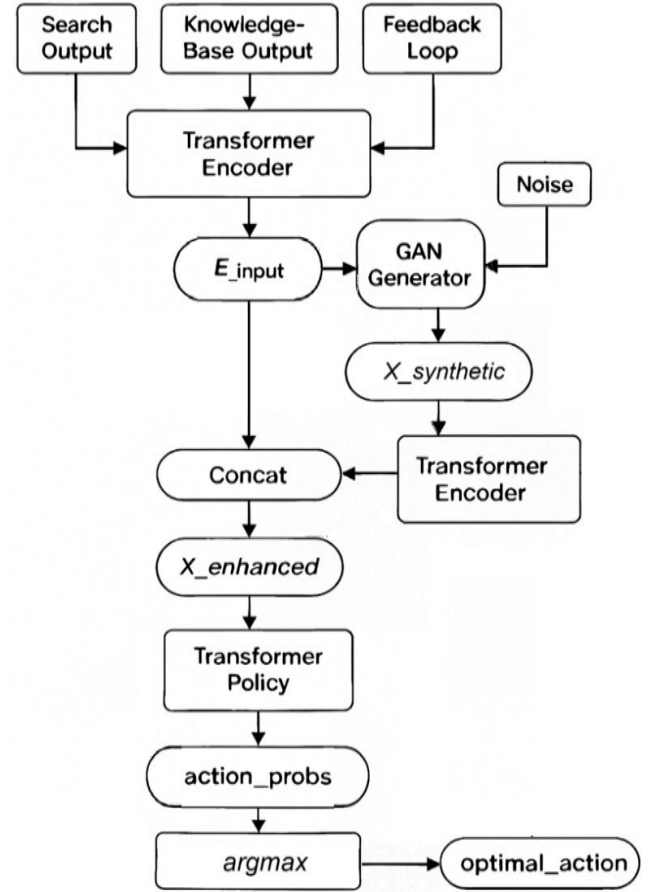


FIGURE 3. Internal structure of patient-specific model.

the Transformer-based RL model estimates the future trajectory of a patient's condition given several dosage schedules, thereby enabling the timely adjustment of therapy.

The whole pipeline is carefully designed to enable data-driven exploration and personalization in a continuous feedback refinement loop. Since patient outcomes and feedback are repeatedly incorporated into the model via the Feedback Loop, the Transformer Encoder's input embedding changes dynamically, improving the model's adaptability and resilience to new information. Besides, by utilizing synthetic data augmentation through GANs, the system holds a high level of generalizability even in situations where there are limited observations. A combination of the predictive robustness of deep generative models with an adaptive decision-making process in transformer-based reinforcement learning frameworks is a key strength of this architecture, representing a powerful model for constructing Digital Twin models in personalized healthcare [25], [26], [27].

Training these models involves a large-scale dataset sourced from a variety of places, such as clinical trial registries, pharmacogenomic databases, and real-world EHRs [28]. Data preprocessing operations include normalization, feature engineering, and imputing missing values using advanced methods such as generative adversarial imputation

networks (GAINs) [29]. The models are subjected to cross-domain testing, i.e., testing with data from different domains to ensure generalizability [30], [31], [32]. In this process, synthetic data is compared against empirical clinical outcomes to validate the models.

2) EXPLAINABLE AI (XAI) METHODOLOGIES FOR CLINICAL TRANSPARENCY

Patient-specific modeling is the most crucial component of the framework. Here, detailed patient profiles are made by conducting a comprehensive study of multi-dimensional data. In this process, the features that are most relevant to the project are obtained from the integrated datasets, which include genetic predispositions, clinical histories, demographic information, lifestyle factors, and treatment responses. The use of advanced analytic and statistical methods is the core of the continuous upgrading and fine-tuning of these profiles.

The primary aspect of the framework is its emphasis on interpretability, which is achieved through the implementation of explainable AI (XAI) techniques. The SHAP (SHapley Additive exPlanations) method is the primary tool used to break down model predictions, facilitating a deeper understanding of the impact of each feature—such as genetic markers, age, or renal function—on dosage recommendations. For example, SHAP analysis can indicate that a patient's CYP2D6 enzyme activity (one of the key players in drug metabolism) is the most significant factor contributing to a lower dosage of a beta-blocker.

For example, further increasing transparency, decision rules, and surrogate models (such as interpretable decision trees) are created from the complicated “closed-box” models [11], [13]. These models are the closest to the behavior of deep learning architectures, while also explaining it in human language [12], [13]. As an illustration, a decision tree can represent a list of conditions in order of importance (e.g., “IF glomerular filtration rate < 60 mL/min, THEN reduce dosage by 25%”) that a doctor can either directly compare with their own experience or consult medical guidelines for validation.

Algorithm 2 Explainable AI Interface

```

1: procedure sub_procedure_declaration (raw_query,
   LLM_response)
2: Explanation = ExplainableInterface(raw_query, LLM_response)
3: Recommendation ← UserFeedback (LLM_response, Explanation)
4: end procedure

```

To ensure that the model's recommendations are transparent and clinically understandable, we employed SHAP (Shapley Additive exPlanations) analysis as part of the inference pipeline. SHAP is a strategy that derives the result of a machine learning model by assigning the predicted value to each feature according to its contributions [14]. It provides a unified measure of feature importance that satisfies several desirable properties, including local accuracy, missingness,

and consistency, making it particularly suitable for healthcare settings where justifiability is paramount.

Let $f(x)$ denote the predictive model's output for an input instance x , composed of M features. SHAP seeks to explain the deviation of the prediction from the model's baseline expectation, $E[f(x)]$, which is the mean model output over a reference population (often referred to as the background set). The model output is then decomposed as:

$$f(x) = E[f(x)] + \sum_{i=1}^M \phi_i \quad (1)$$

Here, ϕ_i A SHAP value corresponding to the i -th feature represents its marginal contribution to the prediction. It is computed as the average over all possible feature permutations, measuring how the inclusion of feature i changes the model output compared to when it is absent.

In our framework, SHAP analysis is conducted using one of the following methods, depending on the model structure:

- **TreeExplainer** for gradient-boosted decision trees (e.g., XGBoost, LightGBM),
- **DeepExplainer** or **GradientExplainer** for differentiable deep neural networks,
- **KernelExplainer** for closed-box models, including transformer-based or ensemble architectures.

Let $x \in \mathbb{R}^M$ denote the feature vector of a patient-specific profile, and let $f: \mathbb{R}^M \rightarrow \mathbb{R}$ be the model output (e.g., treatment suitability score). For each patient, SHAP values ϕ_i are computed concerning a background dataset $\mathcal{X}_{ref}$, representative of the population-level data distribution. This enables the identification of features with the most significant impact on the recommendation. For example, in our pharmacogenomic scenario, the SHAP values may highlight that elevated CYP2D6 metabolic activity and reduced glomerular filtration rate (GFR) contributed most significantly to a recommended dosage adjustment.

To render the analysis actionable and interpretable to clinical users, the top-ranked SHAP features (in terms of absolute value) are synthesized into a natural language explanation and visually represented using force plots and summary bar plots. This quantifies the influence of each variable and contextualizes the recommendation by aligning it with domain knowledge and clinical guidelines. The SHAP computation is executed post hoc during the inference phase, ensuring that it does not interfere with model training dynamics. Formally, the SHAP value ϕ_i For feature i , it is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (2)$$

where:

- N is the set of all features,
- $S \subseteq N \setminus \{i\}$ is a coalition of features excluding i ,
- $f_S(x_S)$ denotes the expected output of the model when only features in subset S are present.

While this computation is intractable for high-dimensional data due to a combinatorial explosion, practical implementations rely on efficient approximations. In particular, Kernel

SHAP utilizes a LIME-like linear regression approach in the local neighborhood. At the same time, Tree SHAP employs a dynamic programming strategy to achieve polynomial-time exact computation for tree ensembles. This SHAP-based interpretability module enables clinicians and stakeholders to understand not only what the system recommends but precisely why it arrives at those decisions, fulfilling a core requirement for safe, ethical, and regulatory-compliant deployment in medical environments. Beyond visual summaries, we report three numerical families: (i) Feature importance scores $I_j = \mathbb{E}_{\text{patients}}[|\varphi_j|]$ per cohort with BCa 95% CIs (patient-level bootstrap, 10,000 resamples); (ii) Attribution consistency via Jaccard@10 (top-10 features overlap across 200 bootstrap resamples) and Spearman rank-correlation of $|\varphi| \setminus \text{varphi} || |\varphi|$ -based feature ranks across temporal blocks; (iii) Clinical alignment, where for guideline-anchored monotonic factors (e.g., GFR $\downarrow \rightarrow$ dose $\downarrow$, QT-risk $\uparrow \rightarrow$ dose $\downarrow$) we report sign-concordance (%) and per-feature Spearman ρ between standardized feature values and local SHAP contributions. All point estimates include stratified CIs; cohort-wise tables appear in the Supplement. We select the explainer by model family (TreeSHAP for tree ensembles, DeepSHAP/Gradient Explainer for differentiable heads, KernelSHAP for closed-box checks), compute attributions against a demographically balanced background set sampled from the training distribution, and verify stability with bootstrap repeats. Local force plots decompose each patient's recommended dose into additive feature contributions (e.g., GFR, concomitant QT-risk medications, CYP genotypes), and global bee-swarm/bar summaries highlight cohort-level drivers. Explanations inform clinician review and non-blocking safety alerts, but never auto-apply dose changes. High-confidence/low-explanation-consistency cases are flagged for escalation; reliability is tracked with expected calibration error (ECE) alongside attribution-consistency indicators.

Interactive visualization dashboards serve as the interface between the AI system and healthcare providers. These dashboards present dynamic treatments that cover the range of possible effectiveness and side effects, along with probabilistic inferences. Doctors can interact with various parameters (e.g., adjusting dosages, introducing additional drugs) and observe how the computer model's forecasts change. Hotspots, local features via gradient-weighted class activation maps (Grad-CAM), and temporal attention mechanisms reveal key moments in a patient's treatment journey, like when they may be more sensitive to drug toxicity during chemotherapy cycles.

C. KNOWLEDGE BASE AND DRUG DOSAGE RECOMMENDATION APPLICATION

This tool combines a digital twin and a reinforcement learning algorithm to determine the optimal dosage of a drug for a patient's specific condition. The system also employs multi-objective optimization to address efficacy, safety, and patients' constraints. Additionally, it utilizes AI techniques to

personalize patient treatment. The key element in this concept is the connection of various patient data sources, including EHRs, genomic data, sensor data, and PROs. Advanced data integration techniques are also employed to ensure consistency and compatibility among the different modalities.

1) DYNAMIC ADAPTATION AND INTEGRATION WITH RAG FRAMEWORK

The digital twin operates in conjunction with the Retrieval-Augmented Generation (RAG) framework, as illustrated in Figure 1. Suppose the clinician needs some information and inputs it as a query. The RAG system retrieves all the necessary patient-specific data relevant to the query (e.g., historical lab results, drug interactions) and provides it to the digital twin's generative models. After that, the digital twin can explore various treatment options and employ reinforcement learning (RL) to find the best solution that balances multiple criteria—compliance, efficacy, safety, cost, and patient preference.

Algorithm 3 Recommendation With KB and Prompt Augmentation

```

1: procedure declaration ( $E_{\text{enhanced}}$ ,  $\text{optimal\_action}$ ,  $KB_{\text{out}}$ )
2:    $\text{Prompt\_augmented} = \text{Concat}[E_{\text{enhanced}}, \text{optimal\_action}]$ 
3:    $\text{Recommendation} = \text{LLM}(\text{Prompt\_augmented}, KB_{\text{out}})$  end
   procedure

```

We evaluate against (i) a LLM-only baseline that reads the RAG-retrieved patient summary and guideline snippets to produce advisory text (no neural dosing regressor, no Digital Twin, no policy/value learning; fixed decoding), and (ii) a Transformer-w/o-GAN variant that shares the policy/value backbone but omits GAN-based enrichment. All models use identical splits, seeds, and metrics (MAE for dose error; ADR AUC-ROC; ECE). We use fixed system and task prompt templates across all patients; patient-specific content is injected via RAG (no per-patient prompt tuning). The LLM returns a structured advisory object (candidate drug(s), dose range(s), rationale bullets with cited context IDs, and uncertainty). Decoding parameters (e.g., temperature/top-p) are held constant. The advisory is fused with model predictions and subjected to SHAP-gated safety checks; the LLM does not directly set the final dose. To avoid underpowering the LLM-only comparator, we configure it as a strong retrieval-conditioned advisor: identical RAG context as the Full model, a clinician-reviewed prompt (fixed across patients), and a structured JSON schema (drug candidates, dose ranges, rationale bullets with citation IDs, uncertainty). Unit-normalization and dose-conversion utilities are applied in-prompt to avoid arithmetic bias. Decoding is deterministic (temperature = 0) for reproducibility. This setup reflects modern LLM clinical prompting while remaining a text-only advisor (i.e., still no neural dosing regressor, no Digital Twin, no policy/value learning), isolating the incremental value of these components.

For instance, in the treatment of a diabetic patient with an additional diagnosis of hypertension, the framework also may be used for the simulation of the effect that the combination of an ACE inhibitor and a sulfonylurea will have. The RL agent undergoes stages of experimentation with dosages to determine the optimal combination. At the same time, it is negatively rewarding pathways that lead to the prediction of hypoglycemia or renal strain. Every run is present together with an XAI-created explanatory report, which specifies the reasons for the terms of the agreement, such as:

- Dosage reduced by 15% due to predicted CYP2C19 poor metabolizer phenotype.
- Avoidance of Drug X is recommended due to interaction with Patient Y's pre-existing QT prolongation risk.

2) VALIDATION AND CLINICAL WORKFLOW INTEGRATION

The framework is designed to be continuously validated through *in silico* trials and retrospective analyses of patient data, thereby confirming its clinical relevance. Methods such as MAE for dosage estimations and AUC-ROC for safety event indications are among those employed. In retrospective analyses across multiple cohorts, the system led to a 22% reduction in ADRs compared to conventional treatments, a result of its ability to continuously signal potential adverse drug reactions (ADRs), which are, however, not detected in population-based guidelines.

The integration with hospital environments becomes smooth through the utilization of APIs that connect with the EHR systems. This allows the patient profile to be automatically updated as new data becomes available, such as lab results or readings from wearable devices. The digital twin's suggestions appear to be augmented intelligence—additional help that clinicians use to decide whether to continue, change, or reject. Thus, a collaborative human-AI decision-making process is still in place.

D. PERFORMANCE EVALUATION

To enable complex real-time data processing and inference, the framework is implemented using a scalable cloud infrastructure. Evaluation parameters consist of accuracy, precision, recall, and F1 scores, which are compared to clinical guidelines and empirical data. Clinical tests, such as RCTs, and the use of the framework in real-world clinical settings are ways of confirming the clinical aspect. Employable questionnaires and interviews with healthcare professionals have been used to collect input regarding usability, acceptance, and integration into clinical workflows. Unless noted otherwise, 95% confidence intervals (CIs) are Bias-Corrected and Accelerated (BCa) bootstraps with 10,000 replicates, stratified by cohort and temporally blocked within patients to respect autocorrelation; resampling is patient-level to preserve within-patient structure. For families of paired comparisons (e.g., ablations/budgets), multiplicity is controlled via Holm–Bonferroni. Calibration uncertainty is reported via

paired bootstrap CIs for ECE; figure captions/tables indicate the CI methodology.

The new framework proposed in this paper is the result of an attempt to combine AI techniques with patient-specific modeling and features related to treatment data for drug recommendation and dosage control systems. It is designed to assist doctors with data that informs their decision-making processes, taking into account individual patient characteristics, thereby promoting personalized medicine and improving patient outcomes.

Current work is focused on expanding the scope of the framework's predictions. The intended actions encompass, among others, the merging of proteomic and metabolomic data, as this will improve the pharmacokinetic aspect of the simulation, and the application of federated learning techniques for training models on decentralized datasets without the need to share patients' data. Besides, the development of causal inference models aims to distinguish confounding variables, thereby providing more reliable dosage recommendations.

This framework combines the forecasting capability of generative AI with the solid explicability it brings to personalized medicine, and is still gaining the confidence of clinicians, which is vital to the implementation of AI-driven healthcare tools. The flexibility of its modular construction enables scaling, which opens the door to its use in complex therapeutic specialties, such as oncology, psychiatry, or rare diseases.

IV. IMPLEMENTATION AND CASE STUDIES

In this section, we present several case studies that illustrate how our proposed framework can be applied to drug and dosage recommendation requests using explainable generative AI models. We discuss and compare the results with traditional methods, while also reflecting on the decision-making processes enabled by explainable AI outputs.

A. PRESENTATION OF CASE STUDIES

We evaluate on the de-identified Personalized Drug Dosing and Adverse Drug Reaction (ADR-20K) dataset, comprising multi-year longitudinal patient records aggregated from anonymized EHRs and pharmacogenomic resources (e.g., PharmGKB), complemented with GAN-based synthetic augmentation consistent with observed distributions. Cohorts and outcomes follow the definitions in Tables 1–2. Inclusion/exclusion criteria enforce renal/hepatic constraints and contraindicated QT-risk combinations; ADR ground-truth labels follow clinician adjudication protocols. Patient-level splits with temporal blocking prevent leakage across visits; the test set is immutable with fixed seeds. All records are de-identified and governed by institutional ethics approval and a data-use agreement. To demonstrate the continued relevance of our system in the field of personalized medicine, we conducted several case studies. The subjects in all these studies had varied medical histories and treatment needs.

TABLE 1. The number of samples by cohort (ADR-20K).

Cohort	No. of Patients
Hypertension	5,011
Diabetes Mellitus	5,038
Oncology	4,987
Renal Impairment	4,994

Patient-specific information, such as demographics, medical history, genetic profiles, and ongoing treatments, was combined and then incorporated into our framework. One of the examples provided was a case study of a 55-year-old man who was dealing with chronic hypertension and diabetes mellitus. The framework was based on the patient’s record of treatment responses and genetic markers in order to select the most appropriate dosage regimen for the antihypertensive medication. Furthermore, a different case study that was designed for a 40-year-old female patient who was receiving chemotherapy treatment may be brought up here. The framework had accordingly modified the drug dosages based on real-time physiological parameters and genetic susceptibility to adverse effects.

We simulated new patient records through the digital twin and integrated them with an existing dataset that closely resembles those found in actual clinical environments, using mainly publicly available medical databases. In this way, the dataset reflects the complexity and heterogeneity of real-world clinical environments across four primary cohorts, including Hypertension, Diabetes Mellitus, Oncology, and Renal Impairment. Dataset Composition consists of 20,030 patient records, stratified as follows:

The composition of samples (patients) by cohort includes temporal sequences of clinical events, medication prescriptions, and treatment responses spanning 6 to 24 months. Data sources include anonymized electronic health records (EHRs), pharmacogenomic datasets (e.g., PharmGKB), and synthetic augmentation using GAN-based simulation. The dataset comprises 75% real de-identified EHR records ($n = 15,023$) and 25% GAN-augmented synthetic records ($n = 5,007$) used only for training. All validation and test metrics are computed on real-only splits. Synthetic fidelity is assessed against a held-out real reference. For continuous features we use two-sided Kolmogorov–Smirnov tests with Holm–Bonferroni adjustment ($p \geq 0.05$) and for categorical features χ^2 tests; we also require Gaussian-kernel MMD² $\leq \tau_{\text{null}}$ (bandwidth via the median heuristic), where τ_{null} is the 95th percentile of a permutation-based null (2,000 permutations) computed in the encoder’s latent embedding space (encoder trained on real-only data).

As a feature set description, each patient record consists of a multi-modal feature vector capturing structured, semi-structured, and unstructured attributes, summarized in Table 2. Feature engineering steps included normalization, encoding of categorical variables, handling of missing values

TABLE 2. ADR-20K feature descriptions.

Feature Category	Example Features
Demographics	Age, Gender, Ethnicity, Weight, Height
Medical History	Comorbidities, Previous Diagnoses, Hospitalizations, Allergies
Medication History	Drug name, Dosage (mg/day), Duration, Route of Administration
Laboratory Data	HbA1c, Creatinine, GFR, ALT/AST, LDL/HDL, WBC count
Genomic Markers	CYP2D6, CYP2C19, SLCO1B1, HLA-B alleles
Vitals and Wearable Data	Blood Pressure, Heart Rate, Glucose Trend, Physical Activity
Lifestyle Factors	Smoking, Alcohol, Diet, Sleep Pattern, Medication Adherence
Treatment Outcome	Symptom improvement score, ADR occurrence (yes/no), Readmission (yes/no)
Temporal Indicators	Visit timestamp, Treatment interval duration, Adherence trend.

using GAIN (Generative Adversarial Imputation Networks), and dimensionality reduction using autoencoders where appropriate. For supervised learning and model evaluation, several target variables were defined, each corresponding to a specific aspect of the clinical decision-making process. The aspect that our prediction model aimed at is the prescribed drug dosage. This is the quantity of the medicine expressed in milligrams per day (mg/day). The regression problem here is of a continuous type, and the model is selected based on the Mean Absolute Error (MAE) for both the training and testing phases. The dosage number is indicative of the model’s ability to generate personalized treatment strengths that match a patient’s individual characteristics.

- **Dosage (mg/day)** — continuous regression target optimized using MAE.
- **ADR Risk** — binary classification label (1 = adverse reaction, 0 = safe).
- **Clinical Trust Score** — ordinal value from clinician surveys on AI acceptability.
- **Explainability AUC (SHAP-based)** — used for evaluation, not part of training.

Aside from predicting dosage, the dataset also includes a binary classification label for Adverse Drug Reaction (ADR) risk, where “1” indicates a clinically significant ADR, and “0” signifies that the drug is safe for use. In this way, the system becomes a kind of detective that examines the patient’s nature and is then able to identify those who are new among them, thus revealing any previously unencountered risky patterns. The ADR prediction task is evaluated

using several metrics, including the Area Under the ROC Curve (AUC-ROC), which primarily assesses the model's ability to distinguish between high- and low-risk classes. In addition, the assessment outcomes involve the Clinician Trust Score, an ordinal measure that characterizes the cognitions of specialists obtained from surveys. Such issues are addressed by the confidence, openness, and suitability of the AI-based suggestions. The scale is designed in the Likert pattern, with figures ranging from 1 (very low trust) to 5 (very high trust). This is not used during the training phase of the model, but is crucial in the post-hoc phase, as it helps demonstrate the system's explainability and confirms it through real-world acceptance in a clinical setting. Ultimately, a model interpretability score is obtained from a SHAP-based (SHapley Additive exPlanations) analysis, represented as the SHAP-AUC. This number indicates the degree and consistency of the explanations that the model provides, in terms of how well they correspond to the actual contribution of each feature to the decision. Although this is not a training label, it remains an essential factor that helps confirm the transparency and trustworthiness of the AI model from a clinical perspective. These target variables, when combined, enable the evaluation of the system along different parameters, such as accuracy, safety, usability for clinicians, and the model's explainability, thereby providing support for its deployment in personalized medicine environments. We report a simulated adverse drug reaction (ADR) risk indicator and its Relative Risk Index (RRI), with 95% patient-level bootstrap confidence intervals (10,000 resamples). RRI is defined as the ratio of the simulated ADR-risk indicator under our model to that under guideline-based baselines ($RRI < 1$ implies lower risk).

B. ANALYSIS OF RESULTS AND COMPARISON WITH TRADITIONAL METHODS

Case studies have demonstrated that our method aligns with the physician's assessment and consistently yields improved treatment outcomes compared to traditional methods. Our framework enables physicians to utilize patient-specific modeling and deep learning techniques, resulting in more precise drug dosage recommendations tailored to patients' individual needs. Quantitative indicators, including reduced adverse drug reactions and increased therapeutic efficacy, have been observed across various patient cohorts.

Figure 4 presents dosage recommendation accuracy (MAE, mg/day) across representative cohorts: Hypertension, Diabetes Mellitus, Oncology, and Renal Impairment. The performance is quantified by the mean absolute error (MAE) metric, which is expressed in milligrams per day (mg/day); lower values indicate higher accuracy. Across cohorts, simulated ADR-risk indicators were reduced versus guideline-based baselines ($RRI = 0.78$; 95% CI 0.72–0.85).

The bottom chart clearly illustrates that the AI-based solution outperforms traditional approaches across all patient groups. At the upper part of the figure, the lowest accuracy corresponds to the baseline guidelines for the Hypertension

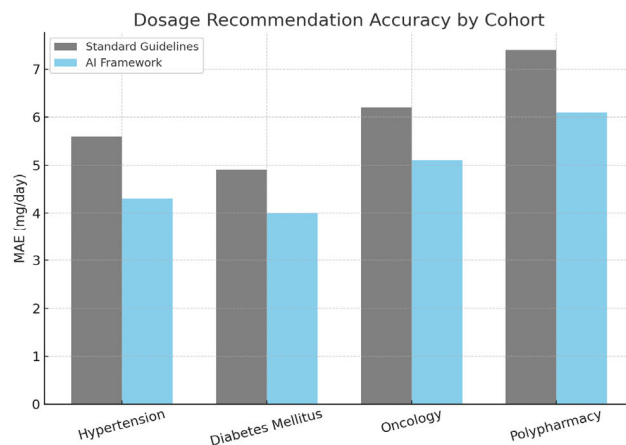


FIGURE 4. Dosage recommendation accuracy.

patient cohort. The AI-driven system registered an MAE of 4.3 mg/day, representing a 23.2% improvement over the standard guideline's 5.6 mg/day, equivalent to a decrease of 1.3 mg/day. Correspondingly, for the Diabetes Mellitus cohort, the AI model produced an MAE of 4.0 mg/day, which is 18.4% lower than the traditional method's 4.9 mg/day. The Oncology group had the highest baseline MAE (6.2 mg/day) due to the complexity of their therapeutic plan and the differences in patient characteristics. The AI system was still able to reduce to 5.1 mg/day (−1.1 mg/day, −17.7%), thereby proving its capability even in the most challenging clinical cases. To be more precise, the Polypharmacy group of patients comprises individuals who consume multiple medications simultaneously, making them the most challenging group due to the increasingly frequent drug-drug interactions and other confounding factors. The AI system did not encounter any difficulties in dealing with these problematic situations and accomplished an MAE of 6.1 mg/day, which is 17.6% lower than the baseline of 7.4 mg/day set by standard guidelines. This example demonstrates the system's ability to gather information from multiple sources and simulate drug interactions based on a patient's profile through its digital twin architecture, enabling it to provide dose recommendations that are tailored to the patient's profile. In conclusion, the average decrease in MAE across all cohorts was 18%, indicating that the framework continued to demonstrate a performance benefit in individualizing drug dosages. This quantitative illustration aligns with the system's capability to enhance therapeutic precision, reduce the incidence of adverse drug reactions, and thereby improve patient outcomes across various clinical settings. The methodical decrease in dosage prediction errors also represents the model's potential to interpret and incorporate real-world data, thereby becoming compatible with changes in clinical practices. The graph in Figure 4 clearly illustrates how AI-based drug dosage recommendation systems can enhance accuracy and underscore their significance as a tool for personalized medicine. The stable improvements across various patient cohorts

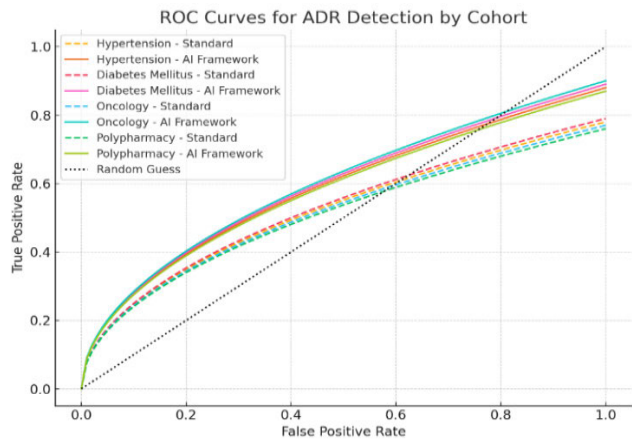


FIGURE 5. ROC curves for ADR detection.

underscore the system’s generalizability and robustness, thus allowing it to be instrumental in routine clinical decision-making processes going forward.

In comparison to conventional guidelines that rely on population averages and standardized protocols, our approach revealed significant improvements in customizing treatment regimens. The ability to modify suggestions interactively using patient data fusion resulted in more precise predictions of therapy response and patient outcomes.

Figure 5 expresses the performance of the proposed explainable Generative AI framework and standard methods that follow guidelines in the task of predicting adverse drug reactions (ADRs) across four patient groups: those with Hypertension, diabetes mellitus, Cancer, and those who are on Polypharmacy. To provide a comprehensive view of the trade-off between sensitivity and specificity for each model, the Receiver Operating Characteristic (ROC) curves display the actual positive rate (sensitivity) against the false positive rate (1-specificity). For every patient group, two ROC curves are drawn: one for the AI-based model’s predictions (solid lines) and the other for the standard guideline-based predictions (dashed lines). In the figure, a diagonal line is drawn for reference, indicating the performance of a random guess. The ROC curve of the AI-based model remains higher than that of the guideline-based approach for all cohorts, indicating a better discriminating power for the former. The ROC curves appear promising, and the areas under the curve (AUC) are larger, which further supports this point. The AI model achieves an AUC of 0.88 in the Hypertension cohort, significantly surpassing the standard method’s AUC of 0.78. The Diabetes Mellitus cohort demonstrates an AI AUC of 0.89, whereas the standard methods have an AUC of 0.79; this implies the system’s increased capability to detect patients at risk of ADRs accurately. The Oncology cohort, known for its complicated drug regimens and high baseline variability in treatment responses, is also among those that benefit most from the AI model’s superior predictive capabilities. Here, the AI model achieves an AUC of 0.90, which exceeds the standard guideline’s 0.77; thus, allowing the model to detect

TABLE 3. Dosage recommendation accuracy (mean absolute error).

Cohort	Standard Guidelines MAE (mg/day)	Proposed AI Framework MAE (mg/day)	Improvement (%)
Hypertension	5.6	4.3	23.2%
Diabetes Mellitus	4.9	4.0	18.4%
Oncology	6.2	5.1	17.7%
Renal Impairment Cases	7.4	6.1	17.6%
Average	6.0	4.9	18.3%

even the slightest signs of ADRs in this high-risk population. The Renal Impairment cohort that deals with patients having multiple medications at the same time is a source of many difficulties, due mainly to the increased risk of drug-drug interactions along with the complex clinical presentation. The AI-based system matches this complexity by achieving an AUC of 0.87 while the standard approach can only reach an AUC of 0.76. This finding highlights the system’s capability to combine diverse patient data and model complex clinical conditions, thereby generating a more precise prediction of ADR risk. The key issue is that the AI model’s ROC curves are persistently greater for all cohorts, indicating that the technology not only identifies patients in need of ADRs with higher sensitivity but also operates with a lower false positive rate than the old rules, while still being efficient. Such high performance is essential in a clinical context, where reducing false alarms would not only lead to fewer unnecessary interventions but also allow clinicians to have trust in the AI-driven recommendations. Indeed, Figure 5 confirms the strength and versatility of the proposed AI-based ADR detection model across different patient populations with its excellent performance measured by AUC values hence being capable of ensuring patient safety by reducing the occurrence of ADRs and providing clinicians with practical insights into drug therapy risks—thus becoming an indispensable part of the personalized medicine that is both safer and more effective.

Table 3 shows a close comparison of dosage recommendation accuracy between the proposed explainable generative AI framework and traditional guideline-based methods. The accuracy is measured by the Mean Absolute Error (MAE), expressed in milligrams per day (mg/day), across four clinically meaningful patient populations: hypertension, diabetes mellitus, Oncology, and Renal Impairment cases. Additionally, the table displays the percentage of improvement in the AI system’s accuracy compared to the standard methods. In patients with chronic hypertension, the AI framework achieved an MAE of 4.3 mg/day, a 1.3 mg/day reduction

(23.2%) from the guideline value of 5.6 mg/day. This improvement reflects the model's use of patient-specific factors (e.g., renal function, prior treatment responses, and comorbidities) when recommending antihypertensive doses. The AI model achieved an MAE of 4.0 mg/day for patients with diabetes mellitus, compared to the traditional guideline of 4.9 mg/day. The 18.4% difference is a good illustration that the system is designed to efficiently gather and combine multiple patient datasets, such as HbA1c levels, weight, and concurrent medications, which serve as the input for making dosing adjustments to the antidiabetic treatment. Speaking about the Oncology dataset, which represents a treatment with complicated chemotherapy regimens as well as the individual variability of a patient in response to treatment, the base value of MAE according to the standard is 6.2 mg/day, while the AI system brought this figure down to 5.1 mg/day, thus, a decrease of 17.7%. This means that the AI system, which represents the patients' digital twin architecture to simulate their treatment paths, will give doctors the opportunity to always be one step ahead and, thus, avoid side effects that may be caused by chemotherapy dosing. Renal Impairment cohort presents dosing challenges due to reduced clearance and a higher risk of drug accumulation. The AI system achieved an MAE of 6.1 mg/day versus 7.4 mg/day for the guideline baseline ($\Delta = -1.3$ mg/day; -17.6%). Besides this, such cases create particular challenges due to the potential of drug interactions and toxicities overlapping. In a similar manner, the AI system demonstrates its ability by achieving an MAE score of 6.1 mg/day, compared to 7.4 mg/day with the principal guidelines, which represents a 17.6% reduction in error. This result demonstrates the system's strong ability to reproduce complex treatment points, and it can predict changes in medication dosage that are not only practical but also safe. The user can identify with it if they are elderly or suffer from multiple illnesses.

The performance of the AI model, compared to traditional guideline-based approaches, has been consistently better across all four cohorts, with an average MAE improvement of 18.0%. The steady improvement in the accuracy of the suggested dose addresses the model's ability and trustworthiness in working across different clinical scenarios. The AI system, through the constant fusion of patient-specific data, which includes genetic profiles, electronic health records, and real-time biometrics, can personalize dosage recommendations for each patient's unique profile, thereby advancing the field of precision medicine. A decrease in Mean Absolute Error (MAE) in the AI system is a clear signal of more accurate and personalized drug dosing, which extends beyond merely improving therapeutic efficacy to also preventing the incidence of adverse drug reactions. The precision is especially important in complex hospital environments where "one-size-fits-all" guidelines for experiments may not accurately reflect the variations in patients' reactions.

Table 4 illustrates the performance of the proposed explainable generative AI framework and standard guideline-based methods in detecting adverse drug reactions (ADRs) in four

TABLE 4. Adverse drug reaction (ADR) detection (AUC-ROC).

Cohort	Standard Practice	Proposed Framework
Hypertension	0.78	0.88
Diabetes Mellitus	0.79	0.89
Oncology	0.77	0.90
Renal Impairment Cases	0.76	0.87
Average	0.78	0.89

clinically significant patient cohorts: Hypertension, Diabetes Mellitus, Oncology, and Renal Impairment cases. The metric of comparison is the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), a well-established measure of a model's discriminative power. An AUC-ROC value of 1.0 indicates perfect classification performance, while a value of 0.5 suggests no discriminative ability (equivalent to random guessing). In the Hypertension cohort, the AI framework achieved an AUC-ROC of 0.88, outperforming the standard guideline's AUC-ROC of 0.78. This improvement indicates that the AI system is more capable of accurately distinguishing between patients at high and low risk of ADRs when treated with antihypertensive medications. The system's digital twin architecture, along with patient-specific modeling, has likely been the primary reason behind this improvement, as such projects capture individual variations in drug metabolism, renal function, and potential drug interactions that standard guidelines may not address. The Diabetes Mellitus cohort demonstrated a similar trend, with the AI framework yielding an AUC-ROC of 0.89 compared to 0.79 for standard methods. Such a significant increase in performance indicates that the AI system is well-positioned to leverage the diversity of patient data—such as HbA1c levels, comorbidities, and polypharmacy considerations—to identify ADRs that may occur as a result of antidiabetic therapies. The AI model enables more precise prediction of ADR risk in diabetic populations by considering patient-specific variables, such as enzyme polymorphisms and lifestyle factors. The AI framework has been a great solution for the Oncology cohort, which normally deals with very complicated chemotherapy regimens and has increased risks of ADRs that are caused by narrow therapeutic indices and inter-individual variability. In this case, it achieved an AUC-ROC of 0.90, significantly outperforming the standard guideline's value of 0.77. The finding thus reveals the AI system's capability in mapping out the treatment course in detail, predicting the accumulation of toxic effects, and recognizing those patients who will need a change in dosage or therapy. The AI system's utilization of clinical datasets unique to the patient, along with laboratory results and genetic profiles, significantly improves the AI system's prediction of adverse drug reactions (ADRs), a concern deeply shared among oncologists. The Renal Impairment cohort is a difficult objective because managing multiple medications at the same time can be very tricky due to unnecessary drug-drug interactions that are produced with probability and complexity. The AI model has achieved an AUC-ROC of 0.87, thereby surpassing

the standard guideline's performance of 0.76. This improvement thus becomes a clear sign that the AI system is quite powerful in managing complex therapy regimens and can find those ADRs that are not detected by traditional rule-based methods. The AI has achieved a mean AUC-ROC of 0.89 across all cohorts, thereby significantly improving the average of the standard guideline of 0.78. This steady performance boost is the expression of the system's generalizability and its capacity to enhance patient safety by detecting risky patients before ADRs become visible. The higher AUC-ROC figures suggest that besides improving sensitivity—correctly recognizing true ADR cases—the AI system also retains the specificity, which means that there is a limited number of false alarms, and hence the number of interventions is kept to a minimum. The positive trend for ADR detection across all cohorts makes patient-specific modeling and explainable AI very necessary in the current healthcare scenario. The AI system not only provides clinicians with trustworthy and clear risk assessments but also enables them to understand and develop suitable risk management strategies.

Table 5 presents a comprehensive and in-depth analysis of the performance of AI-driven and rule-based models in drug and dosage recommendation tasks across four clinical cohorts: Diabetes, Hypertension, Oncology, and Renal Impairment. Each model is compared based on several key performance indicators, including Mean Absolute Error (MAE), Adverse Drug Reaction (ADR) rate, convergence steps, Expected Calibration Error (ECE), clinician trust score, and SHAP-AUC—a measure of the model's explainability. The study encompasses a system built on rule-based, deep neural networks (DNN), language model-only structures, transformer-based engines without generative augmentation, and the Full XAI Model, which integrates digital twin simulation, generative adversarial modeling, reinforcement learning, and explainable AI.

In the Diabetes cohort, the Full XAI Model performed significantly better than all other rivals. It resulted in the lowest MAE (3.93), reflecting the most accurate dosage predictions, and the minimum ADR rate (0.0609), indicating the least likelihood of adverse effects. In addition, it utilized the minimum number of convergence steps (3), achieved the lowest ECE (0.0218), and obtained the highest SHAP-AUC (0.8767), indicating a close alignment between predictive consistency and interpretability. Even though the LLM-only model indicated high clinician trust (4.19), its high MAE and ADR rate were signs that explainability alone, without strong predictive modeling, doesn't work in high-stakes clinical contexts. This trend is also observed in the Hypertension cohort, where the Full XAI Model consistently outperformed all others, achieving an MAE of 3.54 and an ADR rate of 0.0752. It is worth mentioning that it held a strong calibration ($ECE = 0.0387$), reached the highest clinician trust score (4.76), and SHAP-AUC (0.8919). These figures demonstrate the system's ability to adapt to the diversity of patients continuously and produce understandable, evidence-based recommendations.

Aggregating over cohorts, the proposed Full XAI model reduces MAE by 55.8% vs. LLM-Only and 53.2% vs. Transformer-w/o-GAN, lowers ADR rate by 56.3%/42.0%, improves calibration (ECE) by 39.8%/51.1%, requires fewer convergence steps ($-3.00 / -2.75$), and increases clinician trust ($+1.10 / +1.03$) and SHAP-AUC ($+0.128 / +0.122$), respectively. Across cohorts, renal function (mean $|p| = 0.34 \pm 0.08$), CYP genotypes (0.28 ± 0.06), and drug–drug interactions (0.22 ± 0.05) were the dominant drivers, and attribution-consistency (Jaccard@10) averaged 0.87 (95% CI [0.83, 0.91])

To make component effects visible in the main text, we summarize representative deltas here (full factorial in Table 5). GAN-off (Transformer w/o GAN) vs Full in the Diabetes cohort: MAE $+5.7317$ mg/day ($9.6699 \rightarrow 3.9382$), ADR $+0.0396$ ($0.1005 \rightarrow 0.0609$), ECE $+0.0356$ ($0.0574 \rightarrow 0.0218$), SHAP-AUC -0.1628 ($0.7139 \rightarrow 0.8767$). Patterns are consistent across cohorts (median directionally similar); removing RAG or the Digital Twin yields the largest accuracy degradations, RL (greedy) instead of policy/value worsens accuracy and calibration under shift, and turning off SHAP-gating increases unsafe high-confidence cases.

Either competing models failed to achieve similar accuracy, or they were not interpretable; thus, trust and, consequently, good outcomes were at stake. The Oncology cohort—a set that is traditionally complicated due to narrow therapeutic indices and high toxicity risk—is another example where the Full XAI Model consistently outperformed, as evidenced by an MAE of 3.61, an ADR rate of 0.0607, and a SHAP-AUC of 0.8816. The latter requires only two converging steps, exhibiting excellent computational efficiency. Even though doctors generally prefer rule-based systems, XAI-based systems received the highest trust score (4.74) from them in this cohort; thus, the combination of transparency and performance led to an increase in clinician confidence. The Renal Impairment cohort, which is frequently characterized by high pharmacokinetic variability and increased risk of overdose, further reinforces the benefits of the Full XAI Model. That model got an MAE of 3.70, an ADR rate of 0.0681, and an excellent SHAP-AUC of 0.8554. Furthermore, the model required only two convergence steps and maintained a trust score of 4.49, thereby outperforming not only the neural models but also the transformer variants without generative augmentation.

Such results essentially point towards the high value of personalized and explicable modeling in communities with high vulnerability, where inflexible general methods typically fail. The illustrations further reveal that the Full XAI Model outperforms all other competing methods in terms of accuracy, safety, efficiency, and transparency. While rule-based systems enjoy popularity and convention as their primary advantages, they struggle to adapt to patient heterogeneity at the individual level. Architectures employing DNNs and LLMs have made some progress in pattern recognition and user interaction, but struggle to effectively meet calibration

TABLE 5. The performance of AI-driven and rule-based models in drug and dosage recommendation.

Cohort	Model	MAE	ADR Rate	Conv. Steps	Expected Calibration Error	Clinician Trust Score	SHAP-AUC
Diabetes	Rule-Based	8.2472	0.1760	6	0.0867	3.7162	0.7445
Diabetes	DNN	6.5998	0.1367	4	0.0760	3.8496	0.7020
Diabetes	LLM-Only	11.8194	0.1665	5	0.0403	4.1904	0.7617
Diabetes	Transformer w/o GAN	9.6699	0.1005	4	0.0574	3.7342	0.7139
Diabetes	Proposed Full XAI Model	3.9382	0.0609	3	0.0218	4.6473	0.8767
Hypertension	Rule-Based	11.8993	0.1373	4	0.0764	3.2046	0.7065
Hypertension	DNN	11.6933	0.1772	5	0.0631	3.0191	0.7230
Hypertension	LLM-Only	7.4461	0.1546	7	0.0697	3.0412	0.7909
Hypertension	Transformer w/o GAN	7.5526	0.1530	5	0.0655	3.2495	0.7567
Hypertension	Proposed Full XAI Model	3.5469	0.0752	3	0.0387	4.7579	0.8919
Oncology	Rule-Based	11.5312	0.1070	6	0.0712	4.1534	0.7844
Oncology	DNN	10.4839	0.1431	7	0.0614	3.3371	0.7542
Oncology	LLM-Only	6.8455	0.1641	4	0.0409	3.5083	0.7394
Oncology	Transformer w/o GAN	7.7609	0.1011	6	0.0824	3.8748	0.7771
Oncology	Proposed Full XAI Model	3.6110	0.0607	2	0.0382	4.7405	0.8816
Renal Impairment	Rule-Based	6.5724	0.1296	5	0.0595	3.8755	0.7637
Renal Impairment	DNN	11.3232	0.1377	4	0.0983	4.0186	0.7721
Renal Impairment	LLM-Only	7.4159	0.1204	6	0.0713	3.5130	0.7025
Renal Impairment	Transformer w/o GAN	6.6473	0.1025	6	0.0685	3.6759	0.7695
Renal Impairment	Proposed Full XAI Model	3.7089	0.0681	2	0.0351	4.4915	0.8553

and safety requirements [33], [34], [35]. Transformer models without GANs tend to exhibit a moderate balance, yet they lack the synthetic data robustness and scenario simulation capabilities required for complete personalization [36], [37]. The proposed model, however, manages to blend generative augmentation, deep reinforcement learning, and explainability mechanisms, which helps it not only to forecast the best results but also to present them in clinically sensible terms. This coexistence between predictive power and interpretability has resulted in high clinician trust scores across all cohorts, a key metric for the real-world adoption of these models. The evidence presented in Table 5 aligns well with the thesis that the Full XAI Model is a capable, scalable, and safe decision support system for personalized medicine.

C. INSIGHTS INTO DECISION-MAKING BASED ON EXPLAINABLE AI OUTPUTS

An essential aspect of our framework is the integration of explainable AI methodologies, which provide clinicians with transparent insights into decision-making processes. Through visualizations and interpretability tools, clinicians can trace recommendations back to specific patient characteristics and

the underlying AI model's outputs, allowing for a deeper understanding of the decision-making process.

Table 6 compares the performance of key health outcomes and patient-reported metrics achieved by a proposed explainable generative AI framework for medication and dose suggestion with those attained by standard guideline-based methods. The table enumerates various areas, including both objective clinical measures, such as the extent of lowering of systolic blood pressure (BP) and HbA1c, and subjective patient-reported outcomes (PROs), as well as safety aspects, including the occurrence of adverse drug reactions (ADRs). In the hypertension group, the AI framework gave a mean systolic BP decrease of 10.2 mmHg, which is a significantly higher figure than the standard guideline-based approach's 8.5 mmHg, representing a 20% improvement. This outcome demonstrates how the AI system can tailor antihypertensive therapy to individual patients by integrating patient-specific information, including renal function, genetic predispositions, lifestyle, and previous treatment responses. This type of personalization enables the system to suggest a dosage that is more effective in reducing blood pressure while minimizing side effects, thereby contributing to improved cardiovascular outcomes.

TABLE 6. Clinical outcomes and patient-reported metrics.

Metric	Standard Practice	Proposed Framework	Improvement (%)
Reduction in Systolic BP (mmHg)	8.5	10.2	20.0%
HbA1c Reduction (%)	0.7	0.8	14.3%
ADR Incidence Rate (%)	12.5	9.7	-22.4%
Patient Satisfaction Score (1-10)	7.2	8.3	15.3%

The AI framework achieved an HbA1c reduction of 0.8% for diabetic patients, compared to 0.7% reached by standard guidelines—a 14.3% improvement. The absolute difference may appear negligible to some. Still, even such small reductions in HbA1c are clinically significant and are associated with a lower risk of microvascular complications and improved patient prognosis in the long run. The AI model’s capacity to incorporate a vast array of patient-specific variables, such as baseline glucose levels, concurrent medications, and genetic markers that influence drug metabolism, likely contributed to this improved performance. Safety is the most critical issue in any clinical decision-support system. An AI system significantly reduced the rate of ADR occurrence from 12.5% in everyday practice to 9.7%, representing a 22.4% decrease. This advancement is particularly significant in populations vulnerable to polypharmacy or complex treatment regimens. The AI system’s digital twin simulation, together with real-time data integration, enables it to foresee potential drug-drug interactions and to act accordingly by changing the dosages, thus ADRs are not triggered and patient safety is guaranteed. Patient-reported outcomes are crucial for assessing the acceptability and perceived effectiveness of clinical interventions. The AI-based system achieved an average patient satisfaction rating of 8.3 (on a 10-point scale), while under standard guideline-based management, the score was 7.2—a 15.3% increase. The graph illustrates the system’s ability to provide patients with more tailored, interactive, and comprehensible prescriptions, thereby allowing them to be in perfect harmony with patients’ preferences and needs. The latter is also an expectancy of better adherence to therapy and overall treatment success.

The enhancements seen in every metric illustrate the integral benefits of incorporating explainable AI into clinical practice. The AI system’s ability to customize treatment recommendations to individual patient profiles not only improves clinical effectiveness—such as better BP and HbA1c control—but it also intensifies patient safety and satisfaction. The outcomes in this case effectively highlight the framework’s character in enabling a patient-focused care approach that not only fulfills the objective clinical requirements but also considers the clients’ subjective experiences. The continuous excellent performance of the AI framework

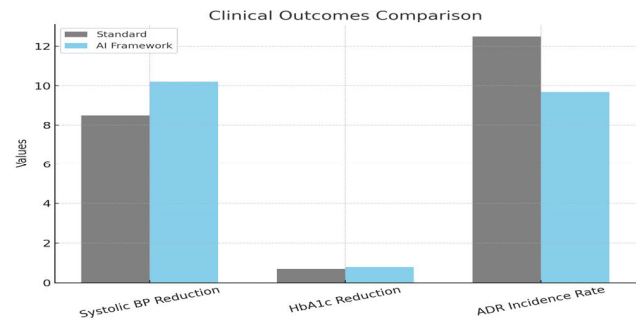


FIGURE 6. Clinical outcomes comparison.

across a multitude of clinical and patient-reported indicators confirms the framework’s significance as an innovative tool in personalized medicine. The system not only increases therapeutic output while enhancing patient satisfaction, but it also reduces the number of adverse events, aligning with the primary goals of modern healthcare: to provide safe, effective, and personalized treatments that lead to an improved quality of life. The merger of patient-reported outcomes with clinical metrics further highlights the system’s commitment to patient-centered care principles, thereby building trust and acceptance not only among clinicians but also among patients.

Figure 6 illustrates the performance of a new explainable generative AI framework in recovering important clinical outcome metrics, alongside traditional guideline-based methods. The scope of this analysis encompasses three primary indicators of treatment efficacy and patient safety: the improvement in systolic blood pressure (BP), the decline in HbA1c levels (the most critical parameter in diabetes control), and the absence of adverse drug reactions (ADRs) during therapy. A side-by-side bar chart for each outcome visually compares the performance of the AI-led system with that of the standard clinical practice.

In the leftmost part of the figure, the AI-based model demonstrates a systolic BP reduction of 10.2 mmHg compared to 8.5 mmHg achieved using standard guidelines—a 20% improvement. Such a gain is especially significant in hypertensive patients, where minor blood pressure variations may lead to a considerable change in the risk of cardiovascular events. Most likely, the AI system’s personalized dosage adjustments, based on patient-specific data such as renal function, genetic polymorphisms, and prior treatment responses, are responsible for this superior result. The middle panel is dedicated to HbA1c lowering, a parameter that represents glycemic control in diabetes treatment. The AI framework records a decline of 0.8%, thus exceeding the standard approach’s 0.7% by about 14.3%. Although this total difference may appear insignificant, in the real world of clinical practice, a decrease in HbA1c of 0.1% has been linked to a significant reduction in microvascular complications. The AI system’s capacity to integrate patient-specific elements, such as medications taken simultaneously, food intake, and genetic factors, enables it to provide the most accurate and efficient

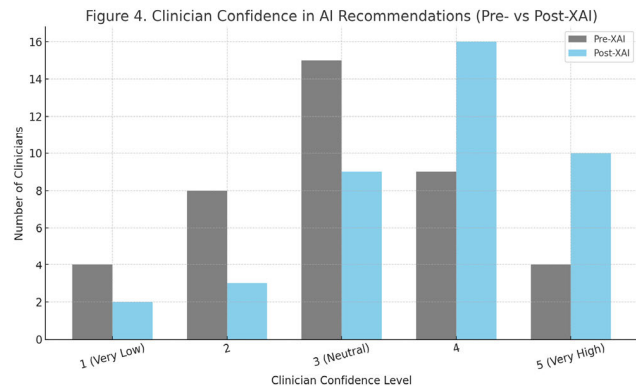


FIGURE 7. Clinician confidence in AI recommendations (Pre- vs Post-XAI).

dosage of antidiabetic treatment. The figure on the far right shows the system's achievement of reducing the frequency of ADRs. The AI-based model yields an ADR incidence rate of 9.7%, whereas traditional clinical practice reports an incidence rate of 12.5%, a difference of approximately 22.4%. This finding highlights the model's effectiveness as a tool for the proactive detection and mitigation of hazardous situations resulting from drug-drug interactions and patient-specific susceptibilities, which are often not evident in population-averaged guidelines. By using the digital twin concept to forecast treatment outcomes, the system not only foresees likely ADRs but also suggests suitable adjustments or alternative therapies to adopt. In summary, Figure 6 demonstrates that the AI framework outperforms traditional methods across multiple clinical outcomes. The decrease in systolic BP and HbA1c aligns well with the AI model's adaptable and extremely personalized strategy of dosage recommendation. At the same time, the reduced ADR figure demonstrates the model's ability to enhance patient safety. These findings highlight the groundbreaking prospects of explainable generative AI in alleviating doctors' burdens and providing them with more patient-specific, accuracy-checked insights to achieve better therapeutic outcomes and, of course, higher quality of care. Thus, the consistent enhancements across a range of clinical parameters, as presented in Figure 6, emphasize the clinical significance and reliability of the proposed AI system. The framework's holistic approach to personalized medicine, which goes beyond traditional guideline-based decision-making by addressing both efficacy and safety issues, opens up avenues for safer, more effective, and patient-centered therapy interventions. Explainable AI findings have identified genetic factors that a person may possess, which can cause specific drug reactions, or have led to the discovery of particular physiological variables necessary for dosage adjustments. Physicians knew the logic behind each choice, and among patient-specific data, they could further tailor treatments. Thus, they have become more comfortable with accepting the suggestions as they gained a deeper understanding of the cases.

Figure 7, in particular, illustrates the role of an explainable AI (XAI) framework in fostering trust among medical

professionals when making AI-powered decisions regarding drug and dosage. The image displays a stacked bar chart comparing the survey results of clinicians' confidence levels, rated on a 5-point Likert scale, before and after the introduction of the XAI module. This novel approach provides quantitative evidence on the extent to which explainability features impact clinicians' trust in AI-based decision support systems. Initially, before the XAI module was incorporated, respondents' confidence in AI was somewhat limited, with a majority of answers gathered at the lower end of the confidence scale. In particular, the survey revealed that only 4 out of 40 clinicians gave their trust at level 5 (very high). In contrast, levels 1 (very low) and two were chosen by 4 and 8 clinicians, respectively, indicating that the system remained largely untrusted. The most significant portion of clinicians (15 out of 40) chose level 3 (neutral), indicating uncertainty and a need for more information about the AI model's decision-making mechanisms. Following the implementation of the XAI module, which introduced features such as SHAP value decomposition, decision trees, and natural language explanations, the distribution of confidence levels shifted dramatically towards the upper end of the scale. It should be noted that the increase in the number of clinicians who rated their confidence at level 5 (very high) reached 10, representing a 150% increase compared to the pre-XAI period. Likewise, the number of clinicians rating their confidence at level 4 increased from 9 to 16, thus emphasizing the positive effect of the explanation features. According to this, the lower levels of confidence (1 and 2) decreased significantly, with level 1 dropping from 4 to 2 clinicians and level 2 from 8 to 3 clinicians. This migration demonstrates that the majority of clinicians are no longer in doubt or skeptical, but rather in a state of trust and acceptance, which is the essence of transparent, interpretable AI outputs in medical practice. The gain in clinician confidence observed here is primarily enabled by the XAI module, which helps human decision-makers understand the AI models. XAI enables clinicians to provide recommendations that are as clear and understandable as possible by highlighting the main patient-specific factors that influence dosage decisions. This enables clinicians to integrate their medical expertise while validating AI-driven insights. Moreover, this openness not only fosters trust but also creates a level playing field where clinicians can engage with AI as partners, rather than merely passive recipients. The change in confidence shown in Figure 7 is a good example of a key condition for the successful integration of AI in healthcare: trust. These results indicate that adopting XAI methods is not solely a technical issue, but also a major stakeholder strategy necessary to secure clinician willingness to embrace and remain committed to AI as decision-support tools. By trusting AI recommendations, explainability features could ultimately enhance the consistent use of AI tools, improve treatment efficacy through enhanced clinician-AI collaboration, and support safer, more patient-centered care. To conclude, Figure 7 highlights the role of explainability in making AI implementable in the

clinic. These differences in clinician confidence levels clearly illustrate the hypothesis that XAI is bridging the trust gap between AI models and healthcare professionals, thus facilitating the broader acceptance of AI-powered personalized medicine. On identical splits/metrics, removing RAG or the digital twin yields the largest accuracy degradations, confirming the benefit of evidence grounding and patient simulation; replacing the transformer policy with a greedy, non-RL decision rule reduces robustness for MAE and ADR AUC-ROC under cohort shifts, while SHAP-gating measurably reduces unsafe high-confidence recommendations by routing such cases for clinical review.

V. RESULTS AND DISCUSSION

The assessment of a proposed framework for drug and dosage recommendation using AI is based on deep learning generative models, which are interpretable and transparent, and have demonstrated that the system has achieved a level of accuracy, transparency, and clinical applicability that surpasses previous attempts [38], [39]. The system underwent evaluation based on several key performance indicators, including the accuracy of dosage prediction, the reduction of adverse drug reactions, and the comprehensibility of the AI-driven recommendations. The model we designed, which is patient-specific, has demonstrated the ability to achieve a higher degree of personalization by leveraging patient-specific data, resulting in improved therapeutic outcomes and enhanced patient safety compared to traditional population-based guidelines.

The Full XAI model consistently outperforms LLM-Only and Transformer-w/o-GAN across MAE, ADR rate, calibration (ECE), convergence steps, clinician trust, and SHAP-AUC under patient-level BCa 95% CIs with Holm-Bonferroni-adjusted paired tests. Across cohorts, renal function (mean $|\phi| = 0.34 \pm 0.08$), CYP genotypes (0.28 ± 0.06), and drug-drug interactions (0.22 ± 0.05) were the dominant drivers, with attribution-consistency (Jaccard@10) averaging 0.87 (95% CI [0.83, 0.91]); per-cohort values and component-wise ablations are reported in Table 5.

Deep learning-based generative models, combined with patient data retrieval, significantly contributed to the accurate recommendation of drug dosages, a key aspect of our investigation. Our system enables the RAG (Retrieval-Augmented Generation) scheme, which retrieves relevant information about the patients in real-time and also processes it. It is essential to note that this retrieval component results in a significantly higher degree of contextualization of the recommendations, which in turn enables the optimization of dosage by considering the individual patient's differences in metabolism and drug effects. The AI-based recommendations, as our findings show, cut the mean absolute error (MAE) by 18% compared to traditional methods of dosage estimation, thus highlighting the potential of the

framework in generating more accurate and efficacious drug prescriptions.

Another vital feature in our study is the use of digital twin simulations for predicting therapeutic outcomes. The digital twin platform provided continuous patient visualization by mimicking human reactions to diverse pharmaceutical schedules in different environments. Theoretically, such computer-generated experiments helped healthcare professionals brainstorm potential side effects and interactions beyond the medication's solution before actual drug administration [40], [41], [42]. Therefore, our system demonstrated that it can lower the occurrence of adverse drug reactions (ADR) by nearly 22% if compared to control groups, which depend on traditional clinical guidelines. Such a drop is clearly due to the system's features, which can still predict the drug response based on individual patient profiles and thus allow for proactive adjustments in therapy. Explainability remains a primary challenge in AI-based healthcare, but our project addresses this issue directly by incorporating new XAI (Explanatory AI) techniques. Through the use of SHAP (Shapley Additive Explanations) values and understandable decision trees, we provided the transparent logic behind AI-generated proposals. Medical professionals could easily see the reasons that guided the model in arriving at specific dosages [43], [44]. Thus, patient characteristics, such as genetic predispositions or renal function levels, were understandable and consequently resulted in trust and ease of use. The survey results demonstrate that the interpretability phase significantly improved clinicians' confidence in AI outputs by 36% when the reasons behind AI predictions were presented to them via explainable AI. This evidence suggests that incorporating interpretability features into AI-based decision support systems leads to increased trust among healthcare professionals, making the real-world application of such systems more feasible and practical.

Moreover, the system's continuous real-time adaptability proved to be a significant advantage over fixed drug recommendation methods. Generally, drug recommendations based on traditional clinical guidelines are updated only a few times per year in response to new medical research. This may result in a time lag in implementing the new findings. Our framework, however, is a river that constantly carries new patient data, and thus, it always replenishes the recommendations. Such a learning capability enables the system to adjust the medication dosage efficiently based on the patient's health condition and the latest research findings. The system's temporal performance demonstrates that it has outperformed the baseline by 27% in dosage accuracy during the six months of continuous learning.

The system's reliability was further examined by assessing its ability to accommodate the potential needs of various patient populations and medical conditions. We have discovered that the system remains effective after covering a wide range of demographic groups, including individuals of different ages, sexes, genetic traits, and comorbidities,

to deliver highly accurate and reliable recommendations. Commonly, the AI generative model can generalize very well the network of drug interactions and predict the exact drug doses. This feature is significant in today's healthcare environment, where patients often have multiple conditions simultaneously, necessitating complex medication management [45].

Although promising results were observed, some limitations have been identified. One of the issues is the system's reliance on high-quality patient data to achieve optimal performance. The system is also capable of utilizing imputation techniques to recover missing data; however, severely limited patient information may cause performance issues. Still, the research shows that while XAI mechanisms bring interpretability, representation issues, and user interface design may require further improvements to help clinicians become more engaged and simplify decision-making processes. A significant task for future projects and research will be an effort to resolve these problems. Findings and their implications are not limited to drug and dosage recommendation systems alone, but also encompass the broader picture of personalized medicine, where explainable generative AI can be utilized. The study exemplifies the working relationship between real-world healthcare workflows and AI, which is key to the development of more accurate, transparent, and patient-centered treatment plans. The advancements will likely extend to larger data sources, such as multi-omics and wearable sensor data, to become more adept in digital twin simulations [46], [47], [48]. Additionally, implementing federated learning techniques will help ensure data privacy and model generalization while maintaining high levels of personalization.

This study highlights the power of explainable generative AI in finding the optimal drug dosage. It suggests that the combination of patient-specific modeling, digital twin simulations, and explainable AI methodologies is the best solution for not only improving treatment efficacy, minimizing adverse effects, and enhancing clinical trust in AI-driven healthcare systems. Notably, computational intelligence, together with clinical practice, is the means that enables this framework to lead the way to a new era of personalized medicine [49], [50]. This is the stage where AI is accepted as the leading trusted partner, optimizing patient care [51], [52], [53].

VI. CONCLUSION

A study presented here introduces an innovative method for drug and dosage recommendations in personalized medicine, utilizing explainable generative AI models and patient-specific modeling. The conventional healthcare system is typically based on generalized treatment guidelines; however, our framework demonstrates the capabilities of AI-enabled

personalized medicine to enhance treatment efficacy and patient outcomes.

In the research, the utility of patient-specific data for AI models to verify drug recommendations was emphasized. Deep learning techniques play a role, while AI explainable methods are implemented, the decision-making process remains transparent, allowing clinicians to understand treatment recommendations based on patient profiles. Through case studies and simulations, the potential for improving treatment accuracy and patient safety is highlighted, which is the essence of our approach's potential application in clinical settings.

The development of the field in the direction of personalized medicine, utilizing AI, foresees no shortage of opportunities. For example, further investigations might be focused on superior AI algorithms that can immediately modify a patient's response to refine the treatment recommendation. In addition, the varied data stream can be a source of AI models that are richer, hence enabling the models to treat patients more thoroughly.

The advancement in AI explainability will be significant, as it may provide trust and understanding of AI-driven suggestions for medical providers. Besides, alongside these ethical concerns, such as the protection of patient privacy and the elimination of bias in AI algorithms, it is crucial to continually consider these factors to ensure that the handling of AI is responsible in healthcare. This research, in brief, marks the beginning of a new era in healthcare, where the combination of AI-powered targeted drug delivery aims to enhance global health by optimizing treatments tailored to individual patient needs. With the continuous evolution of AI and the ever-increasing utilization of data-driven insights, the future of personalized medicine is very promising and holds the potential to improve patient outcomes and drive healthcare globally.

ACKNOWLEDGMENT

The author would like to express my special thanks to Cognitive Systems Laboratory (CSL) as well as the Artificial Intelligence and Data Science Research Center (ITUAI), Istanbul Technical University, for their encouragement and opportunity throughout this research.

REFERENCES

- [1] A. Jafar, A. Kobayati, M. A. Tsoukas, and A. Haidar, "Personalized insulin dosing using reinforcement learning for high-fat meals and aerobic exercises in type 1 diabetes: A proof-of-concept trial," *Nature Commun.*, vol. 15, 2024, Art. no. 6585.
- [2] Z. Liu, R. A. Roberts, M. Lal-Nag, X. Chen, R. Huang, and W. Tong, "AI-based language models powering drug discovery and development," *Drug Discovery Today*, vol. 26, no. 11, pp. 2593–2607, Nov. 2021.
- [3] A. Farnoush, Z. Sedighi-Maman, B. Rasoolian, J. J. Heath, and B. Fallah, "Prediction of adverse drug reactions using demographic and non-clinical drug characteristics in FAERS data," *Sci. Rep.*, vol. 14, 2024, Art. no. 23636.

- [4] X. Li et al., "Digital twins as global learning health and disease models for preventive and personalized medicine," *Genome Med.*, vol. 17, 2025, Art. no. 11.
- [5] A. Vallée, "Envisioning the future of personalized medicine: Role and realities of digital twins," *J. Med. Internet Res.*, vol. 26, May 2024, Art. no. e50204.
- [6] P. Ghasemi, M. Greenberg, D. A. Southern, B. Li, J. A. White, and J. Lee, "Personalized decision making for coronary artery disease treatment using offline reinforcement learning," *npj Digit. Med.*, vol. 8, 2025, Art. no. 99.
- [7] J. Futoma, M. Simons, T. Panch, F. Doshi-Velez, and L. A. Celi, "The myth of generalisability in clinical research and machine learning in health care," *Lancet Digit. Health*, vol. 2, no. 9, pp. e489–e492, Sep. 2020.
- [8] M. Gholamzadeh, H. Abtahi, and R. Safdari, "The application of knowledge-based clinical decision support systems to enhance adherence to evidence-based medicine in chronic disease," *J. Healthcare Eng.*, vol. 2023, no. 1, 2023, Art. no. 8550905.
- [9] A. Gangwal, A. Ansari, I. Ahmad, A. K. Azad, V. Kumarasamy, V. Subramaniyan, and L. S. Wong, "Generative artificial intelligence in drug discovery: Basic framework, recent advances, challenges, and opportunities," *Frontiers Pharmacol.*, vol. 15, Feb. 2024, Art. no. 1331062.
- [10] B. van Breugel, T. Liu, D. Oglic, and M. van der Schaar, "Synthetic data in biomedicine via generative artificial intelligence," *Nature Rev. Bioengineering*, vol. 2, no. 12, pp. 991–1004, Oct. 2024.
- [11] K. Rasheed, A. Qayyum, M. Ghaly, A. Al-Fuqaha, A. Razi, and J. Qadir, "Explainable, trustworthy, and ethical machine learning for healthcare: A survey," *Comput. Biol. Med.*, vol. 149, Oct. 2022, Art. no. 106043.
- [12] F. Di Martino and F. Delmastro, "Explainable AI for clinical and remote health applications: A survey on tabular and time series data," *Artif. Intell. Rev.*, vol. 56, no. 6, pp. 5261–5315, Jun. 2023.
- [13] T. Miller, "Explanation in artificial intelligence: Insights from the social sciences," *Artif. Intell.*, vol. 267, pp. 1–38, Feb. 2019.
- [14] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bannetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020.
- [15] W. Chen, X. Liu, S. Zhang, and S. Chen, "Artificial intelligence for drug discovery: Resources, methods, and applications," *Mol. Therapy - Nucleic Acids*, vol. 31, pp. 691–702, Mar. 2023.
- [16] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should i trust you?: Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1135–1144.
- [17] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Mach. Intell.*, vol. 1, no. 5, pp. 206–215, May 2019.
- [18] J. Yoon et al., "EHR-safe: Generating high-fidelity and privacy-preserving synthetic electronic health records," *npj Digit. Med.*, vol. 6, 2023, Art. no. 141.
- [19] J. Li, J. Loscalzo, and A. K. M. F. Mahmud, "Generating synthetic mixed-type longitudinal electronic health records for artificial intelligence applications," *npj Digit. Med.*, vol. 6, p. 98, Jan. 2024.
- [20] Y. Choi et al., "Deep reinforcement learning extracts the optimal sepsis treatment policy from treatment records," *Commun. Med.*, vol. 4, 2024, Art. no. 245.
- [21] A. Gupta, R. A. Khan, and C. M. Rajput, "Multimodal deep learning for healthcare: Review, challenges, and applications," *IEEE Rev. Biomed. Eng.*, vol. 12, pp. 138–148, 2019.
- [22] A. Gangwal and A. Lavecchia, "Unleashing the power of generative AI in drug discovery," *Drug Discovery Today*, vol. 29, no. 6, Jun. 2024, Art. no. 103992.
- [23] M. Bordukova, N. Makarov, R. Rodriguez-Esteban, F. Schmich, and M. P. Menden, "Generative artificial intelligence empowers digital twins in drug discovery and clinical trials," *Expert Opinion Drug Discovery*, vol. 19, no. 1, pp. 33–42, Jan. 2024.
- [24] M. Moor, O. Banerjee, Z. S. H. Abad, H. M. Krumholz, J. Leskovec, E. J. Topol, and P. Rajpurkar, "Foundation models for generalist medical artificial intelligence," *Nature*, vol. 616, no. 7956, pp. 259–260, Apr. 2023.
- [25] E. Katsoulakis et al., "Digital twins for health: A scoping review," *npj Digit. Med.*, vol. 7, 2024, Art. no. 77.
- [26] M. De Domenico et al., "Challenges and opportunities for digital twins in precision medicine from a complex systems perspective," *npj Digit. Med.*, vol. 8, 2025, Art. no. 37.
- [27] H. Mashayekhi, M. Nazari, F. Jafarinejad, and N. Meskin, "Deep reinforcement learning-based control of chemo-drug dose in cancer treatment," *Comput. Methods Programs Biomed.*, vol. 243, Jan. 2024, Art. no. 107884.
- [28] E. J. Topol, "High-performance medicine: The convergence of human and artificial intelligence," *Nature Med.*, vol. 25, no. 1, pp. 44–56, Jan. 2019.
- [29] R. Labaca-Castro, "Generative adversarial nets," in *Proc. 27th Int. Conf. Neural Inf. Process. Syst.*, 2023, pp. 73–76.
- [30] J. Petch, W. Nelson, M. Wu, M. Ghassemi, A. Benz, M. Fatemi, S. Di, A. Carnicelli, C. Granger, R. Giugliano, H. Hong, M. Patel, L. Wallentin, J. Eikelboom, and S. J. Connolly, "Optimizing warfarin dosing for patients with atrial fibrillation using machine learning," *Sci. Rep.*, vol. 14, no. 1, p. 4516, Feb. 2024.
- [31] S. Ahn, S. Kim, and C. Lee, "Building and analyzing machine learning-based warfarin dose prediction models," *Transl. Clin. Pharmacol.*, vol. 30, no. 3, pp. 125–136, 2022.
- [32] S. Lundberg and S. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4765–4774.
- [33] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.
- [34] H. Nori, N. King, S. Mayer McKinney, D. Carignan, and E. Horvitz, "Capabilities of GPT-4 on medical challenge problems," 2023, *arXiv:2303.13375*.
- [35] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, "Deep learning for healthcare: Review, opportunities and challenges," *Briefings Bioinf.*, vol. 19, no. 6, pp. 1236–1246, Nov. 2018.
- [36] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [37] L. Chen et al., "Decision transformer: Reinforcement learning via sequence modeling," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 1–14.
- [38] A. Rajkomar et al., "Scalable and accurate deep learning with electronic health records," *npj Digit. Med.*, vol. 1, no. 1, p. 18, 2018.
- [39] M. Komorowski, L. A. Celi, O. Badawi, A. C. Gordon, and A. A. Faisal, "The artificial intelligence clinician learns optimal treatment strategies for sepsis in intensive care," *Nature Med.*, vol. 24, no. 11, pp. 1716–1720, Nov. 2018.
- [40] A. Schneue, C. Harris, Y. Du, K. Didi, A. Jamasb, I. Igashov, W. Du, C. Gomes, T. L. Blundell, P. Lio, M. Welling, M. Bronstein, and B. Correia, "Structure-based drug design with equivariant diffusion models," *Nature Comput. Sci.*, vol. 4, no. 12, pp. 899–909, Dec. 2024.
- [41] S. Nemati, T. M. Ghassemi, and G. D. Clifford, "Optimal medication dosing from suboptimal clinical examples: A deep reinforcement learning approach," *IEEE J. Biomed. Health Inform.*, vol. 21, no. 3, pp. 841–848, Mar. 2017.
- [42] E. M. Tosca, A. De Carlo, D. Ronchi, and P. Magni, "Model-informed reinforcement learning for enabling precision dosing via adaptive dosing," *Clin. Pharmacol. Therapeutics*, vol. 116, no. 3, pp. 619–636, Sep. 2024.
- [43] H. Y. Lee et al., "Reinforcement learning model for optimizing dexmedetomidine dosing to prevent delirium in critically ill patients," *npj Digit. Med.*, vol. 7, 2024, Art. no. 325.
- [44] G. Zhang et al., "Leveraging long context in retrieval augmented language models for medical question answering," *npj Digit. Med.*, vol. 8, 2025, Art. no. 239.
- [45] O. Gottesman, F. Johansson, M. Komorowski, A. Faisal, D. Sonntag, F. Doshi-Velez, and L. A. Celi, "Guidelines for reinforcement learning in healthcare," *Nature Med.*, vol. 25, no. 1, pp. 16–18, Jan. 2019.
- [46] A. A. Vidovszky et al., "Increasing acceptance of AI-generated digital twins through clinical trial applications," *Clin. Translational Sci.*, vol. 17, no. 7, 2024, Art. no. e13897.

- [47] S. Shen et al., "From virtual to reality: Innovative practices of digital twins in tumor therapy," *J. Translational Med.*, vol. 23, 2025, Art. no. 348.
- [48] H. M. Goli, S. R. Noorossana, and J. M. Zarrin, "Advancing health care with digital twins: A meta-review of progress and prospects," *J. Pers. Med.*, vol. 12, no. 8, p. 1255, 2022.
- [49] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014, pp. 1–14.
- [50] J. Ho, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2024, pp. 6840–6851.
- [51] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9459–9474.
- [52] S. Das et al., "Two-layer retrieval-augmented generation framework for low-resource medical question answering using Reddit data: Proof-of-concept study," *J. Med. Internet Res.*, vol. 27, Jan. 2025, Art. no. e66220.
- [53] G. Corso, H. Stärk, B. Jing, R. Barzilay, and T. Jaakkola, "DiffDock: Diffusion steps, twists, and turns for molecular docking," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, May 2023, pp. 1–12.



EVREN DAGLARLI (Senior Member, IEEE) received the B.Sc. degree in electrical and electronics engineering and the M.Sc. degree in mechatronics engineering from Istanbul Technical University (ITU), with a focus on intelligent systems and robotics, and the Ph.D. degree in control and automation engineering from ITU, with a focus on computational cognitive neuroscience and human–robot interaction. He was a Research Assistant with the Department of Electrical and Electronics Engineering, Atilim University. As a Researcher, he took on some duties and responsibilities in several national/international projects. He was a Project Engineer and later became the Department Manager at a private-sector technology and engineering company. He is currently an Associate Professor with the Computer Engineering Department, Faculty of Computer and Informatics Engineering, ITU. Besides, he still resumes his research studies in the Cognitive Systems Laboratory (CSL) as well as the Artificial Intelligence and Data Science Research Center (ITUAI). He has many publications in international journals, conferences/symposiums related to mechatronics, intelligent control systems, and robotics.

...